Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 5 Marks

CODING & SOLUTION TEMPLATE

Solution Summary Table

Attach your code repository link and outline your programming environment setup details below:

Field Name	**Details / Specifications**
Repository Link / URL	https://github.com/MaheshKona7/credit-card-approval-v2
Programming Language(s)	Python (3.9+)
Framework(s) Used	Flask (2.0.2), Gunicorn (production WSGI server)
Key Features Implemented	- Layer 1 Policy overrides (4 risk filters) - Stratified join ETL - SMOTE minority-class balancing - Random Forest Classifier with threshold calibration - Responsive dark-theme outcomes dashboard
Pending / Incomplete Features	SHAP/LIME feature contribution explanation charts (planned for v2.1)
Setup / Run Instructions	1. Run: pip install -r requirements.txt 2. Run: python model.py (trains model and serializes pickles) 3. Run: python app.py (starts local server at 127.0.0.1:5000)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes:

The code is optimized for local execution compliance. XGBoost is bypassed because corporate computer policies block external C-DLL compilation, and scikit-learn's Random Forest classifier is used as a fully functional and highly performant alternative.